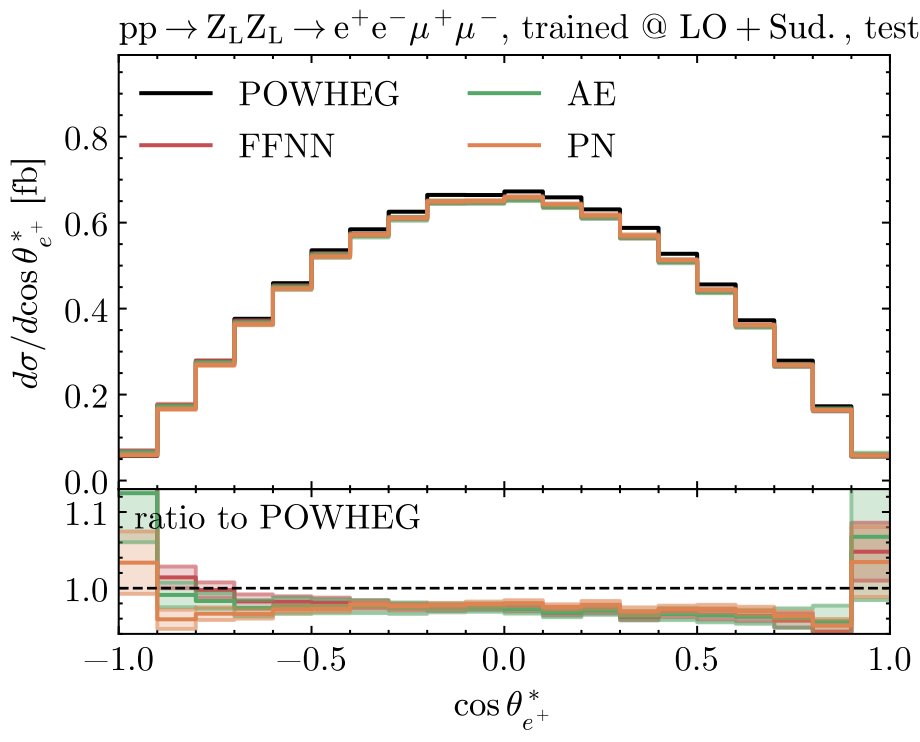
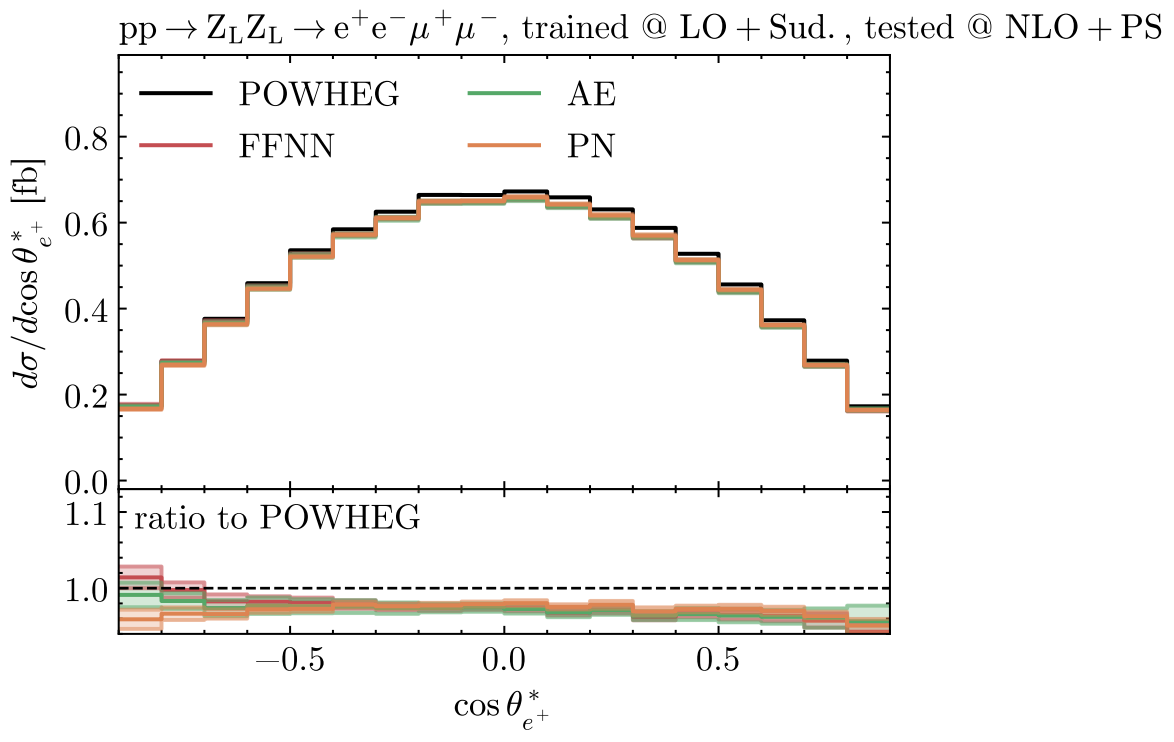
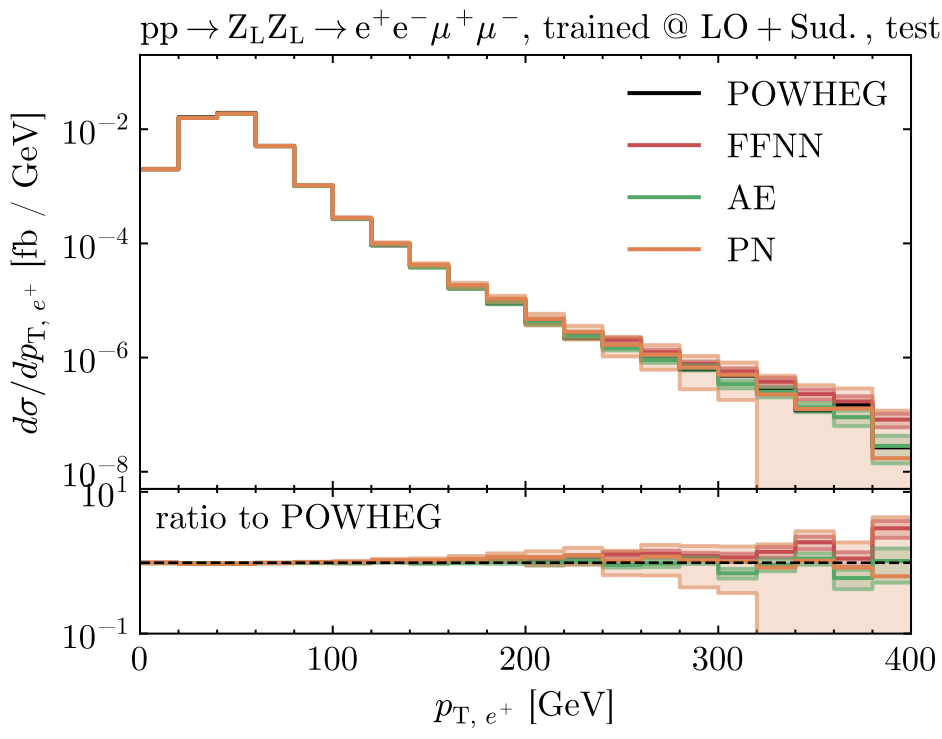
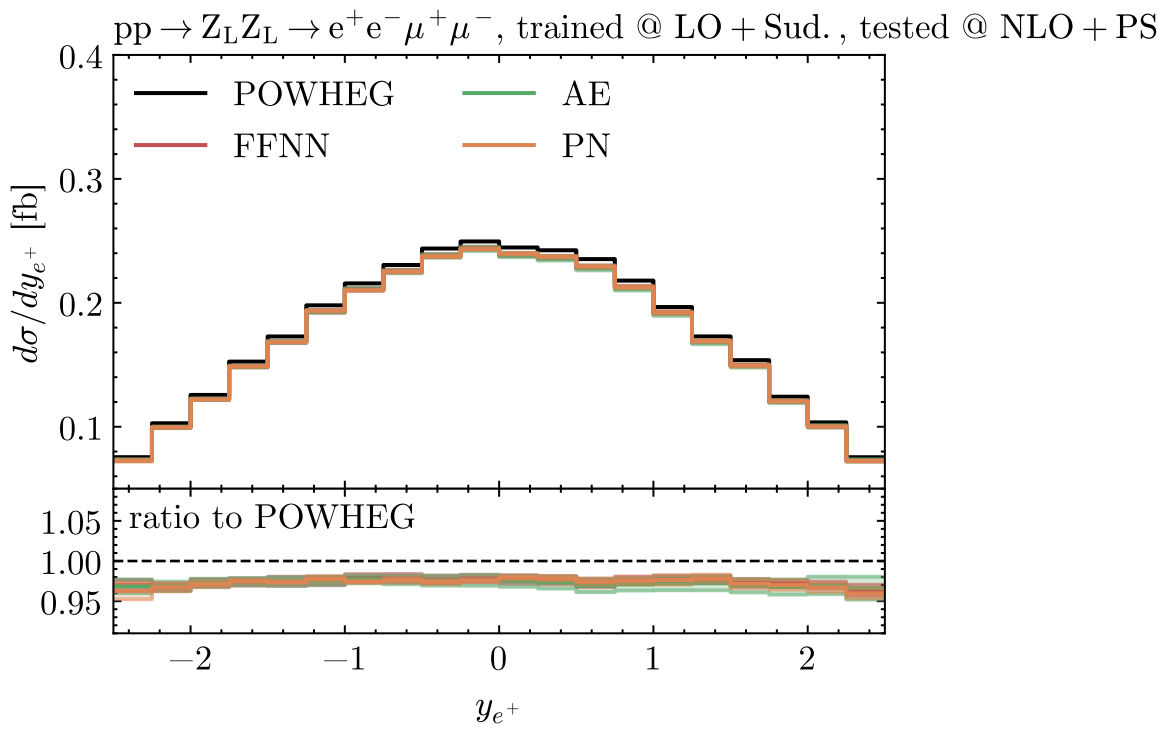
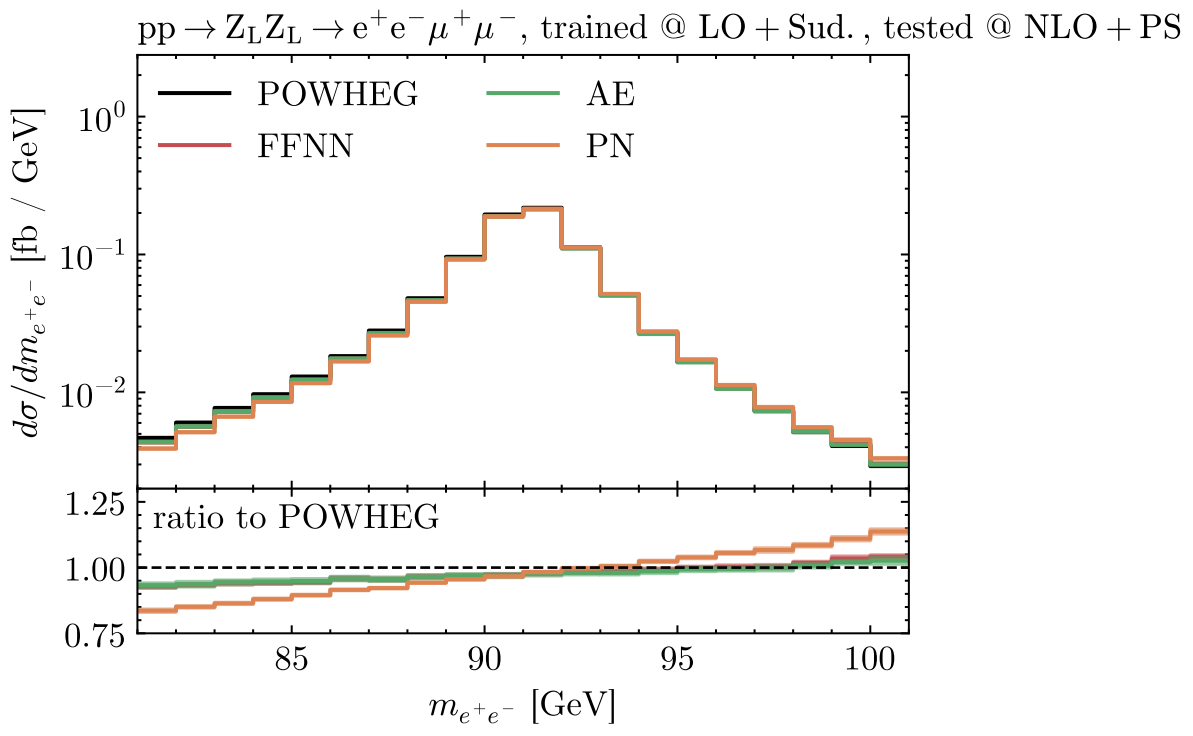
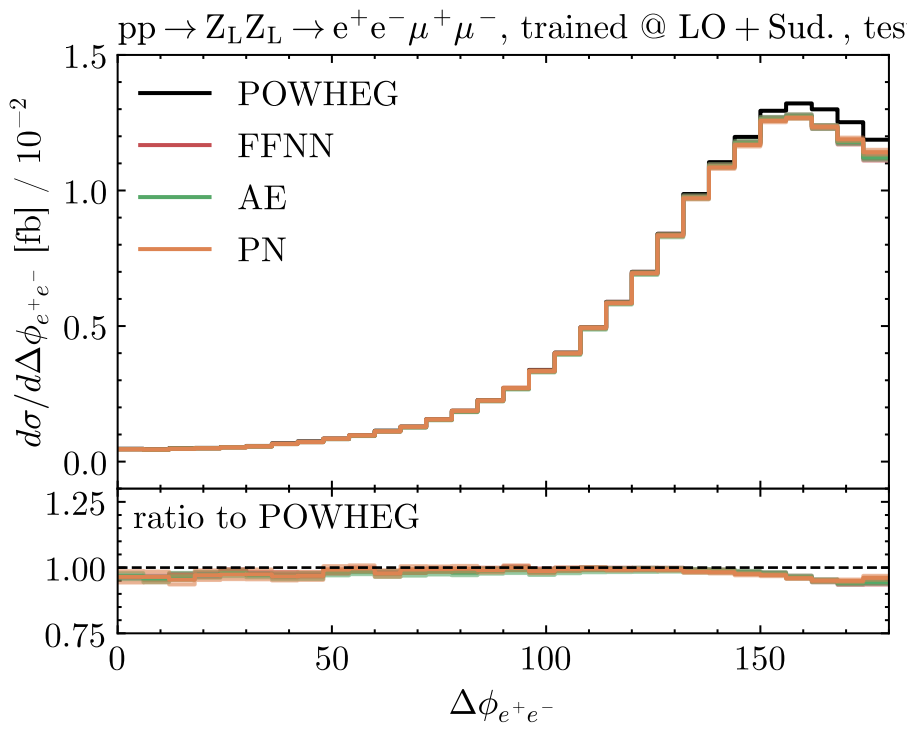

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−1.0
−0.5
0.0
0.5
1.0
$\cos\theta^*_{e^+}$

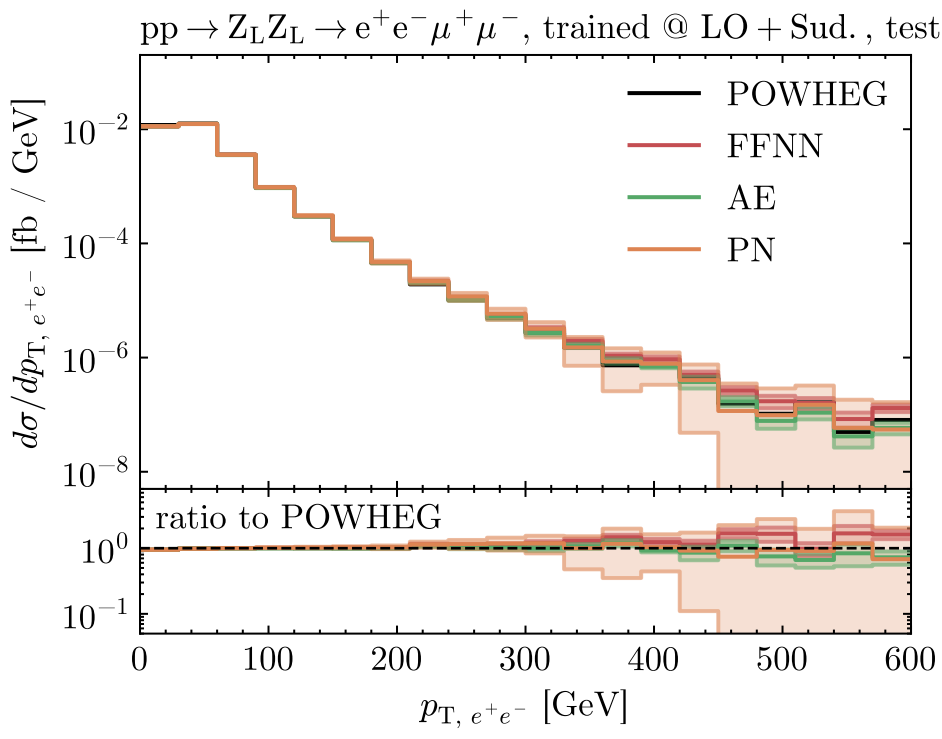

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
$d\sigma/d\cos\theta^*_{e^+}$ [fb]
0.0
0.2
0.4
0.6
0.8
ratio to POWHEG
1.0
1.1
−0.5
0.0
0.5
$\cos\theta^*_{e^+}$

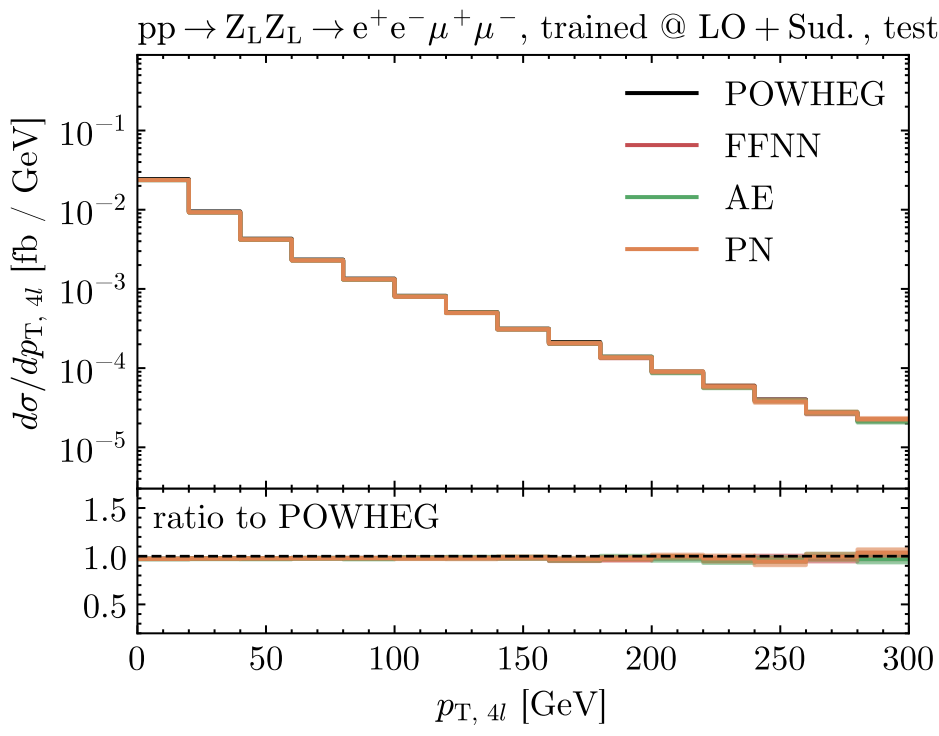

$pp \to Z_L Z_L \to e^+ e^- \mu^+ \mu^-$, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
$d\sigma/dp_{T,\,e^+}$ [fb / GeV]
10^{-2}
10^{-4}
10^{-6}
10^{-8}
ratio to POWHEG
10^{1}
10^{-1}
0
100
200
300
400
$p_{T,\,e^+}$ [GeV]

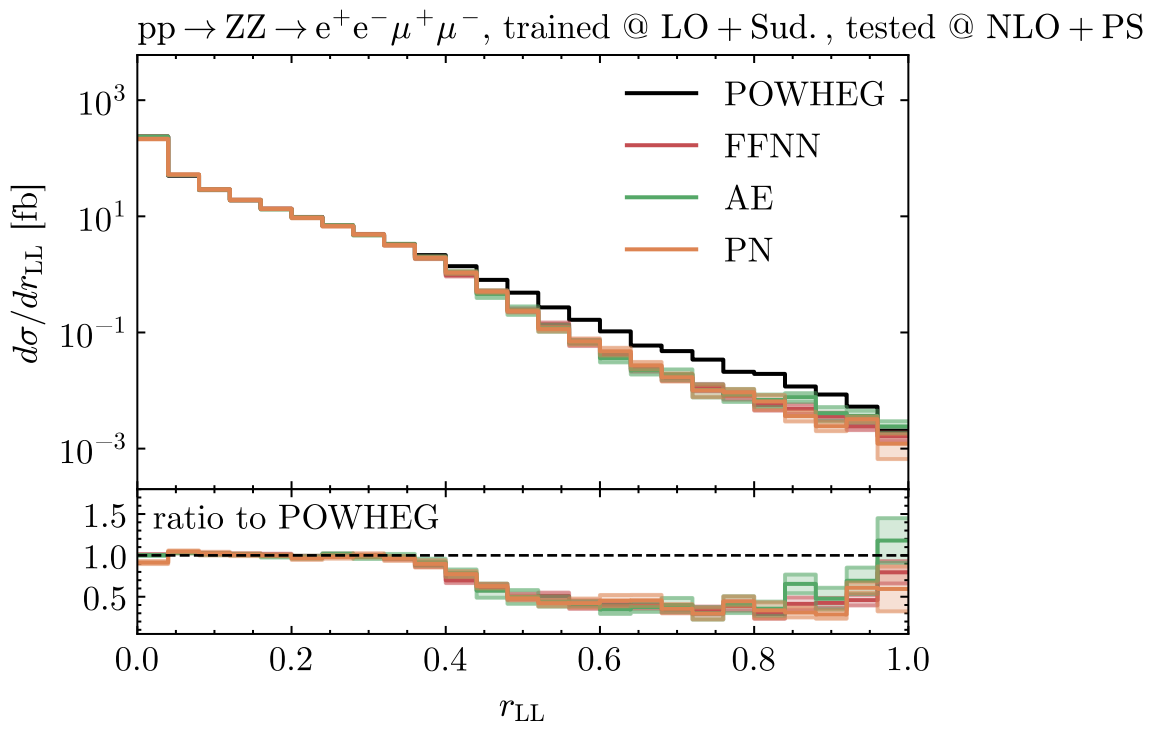

$\mathrm{pp} \to \mathrm{Z_L Z_L} \to \mathrm{e^+ e^- \mu^+ \mu^-}$, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
$d\sigma/dy_{e^+}$ [fb]
0.4
0.3
0.2
0.1
ratio to POWHEG
1.05
1.00
0.95
−2
−1
0
1
2
y_{e^+}

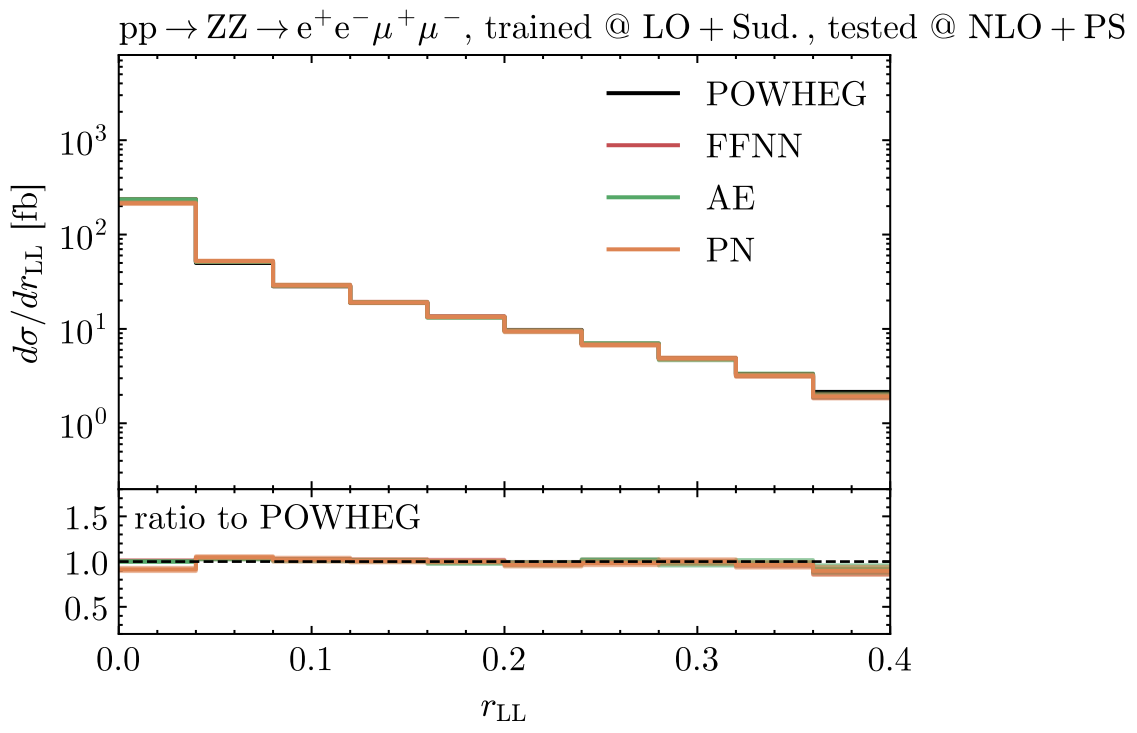

pp → $Z_L Z_L$ → $e^+e^-\mu^+\mu^-$, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
$d\sigma/dm_{e^+e^-}$ [fb / GeV]
10^0
10^{-1}
10^{-2}
ratio to POWHEG
1.25
1.00
0.75
85
90
95
100
$m_{e^+e^-}$ [GeV]


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
dσ/dΔφ_{e⁺e⁻} [fb] / 10⁻²
1.5
1.0
0.5
0.0
ratio to POWHEG
1.25
1.00
0.75
0
50
100
150
Δφ_{e⁺e⁻}


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
dσ/dp_T, e⁺e⁻ [fb / GeV]
ratio to POWHEG
p_T, e⁺e⁻ [GeV]


pp → Z_L Z_L → e⁺e⁻μ⁺μ⁻, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
dσ/dp_T, 4l [fb / GeV]
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
ratio to POWHEG
1.5
1.0
0.5
0
50
100
150
200
250
300
p_T, 4l [GeV]


pp → ZZ → e⁺e⁻μ⁺μ⁻, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
dσ/dr_LL [fb]
ratio to POWHEG
r_LL


$pp \to ZZ \to e^+e^-\mu^+\mu^-$, trained @ LO + Sud. , tested @ NLO + PS
POWHEG
FFNN
AE
PN
$d\sigma/dr_{LL}$ [fb]
10^3
10^2
10^1
10^0
ratio to POWHEG
1.5
1.0
0.5
0.0
0.1
0.2
0.3
0.4
r_{LL}